

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023 - 2024 (ODD Semester)

INNOVATIVE TEACHING METHOD PLAN

Degree, Semester & Branch: III Semester B.Tech.IT.

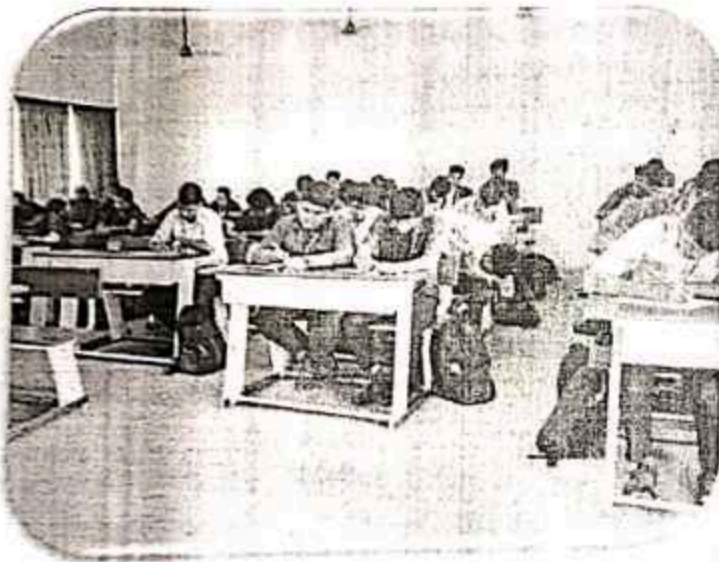
Course Code & Title: CS3351 - Digital Principles and Computer Organization

Name of the Faculty member: Ms.S.Harini Shriram

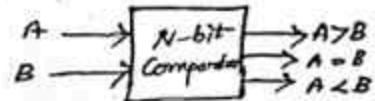
UNIT I

1. 23.9.2023 Magnitude Comparator

One Minute Paper



Magnitude Comparator :-
 ↳ Has 2 input &
 produces 3 outputs.



A comparator used to
 compare two binary numbers

1 Bit Magnitude Comparator

A	B	A < B	A = B	A > B
0	0	0	1	0
0	1	1	0	0
1	0	0	0	1
1	1	0	1	0

Signature of the Faculty Member

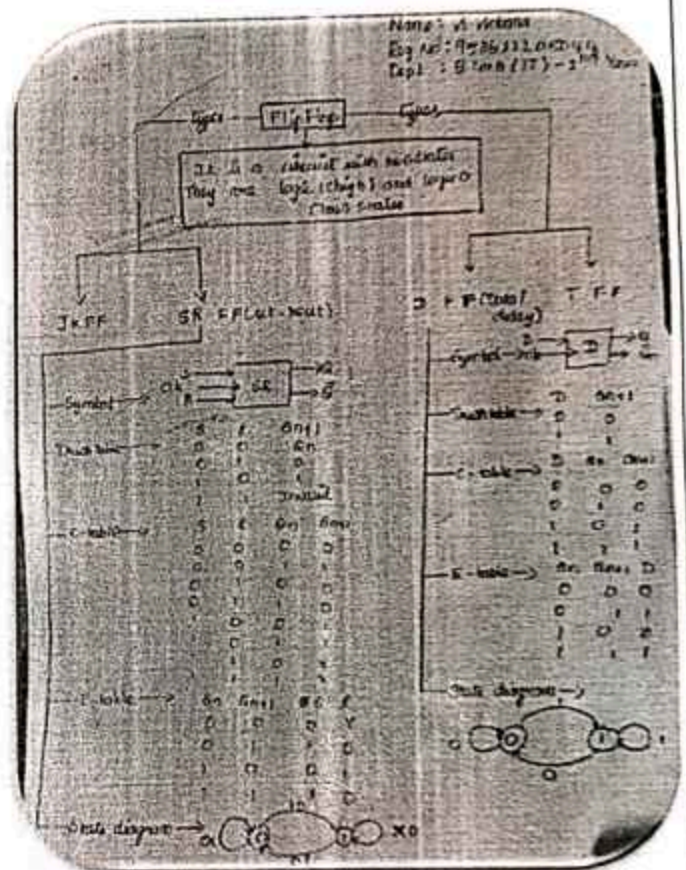
HOD

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1.	6.10.2023	Flip Flops (SR & D) Operation, Excitation Table
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UNIT III

1.

1.11.2023

Von Neumann Architecture &
Operations and Operands

Write Pair Share




1/11/23

Signature of the Faculty Member


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**Department of Electronics and Communication Engineering**
Academic Year 2023 – 2024 (Odd Semester)**Degree, Semester & Branch:** : III Semester B.Tech.IT**Course Code & Title:** CS3351 Digital Principles and Computer Organization**Name of the Faculty member (s):** Ms.S.Harini Shriram**Innovative Practice Description**

- **Unit / Topic:** V / Memory and I/O
- **Course Outcome:** CO : The students will be able Identify the characteristics of various memory systems and I/O communication
- **Topic Learning Outcome:** TLO : To explain the different types of memories and its mapping
- **Activity Chosen:** Jigsaw
- **Justification:**
 - The Jigsaw technique is a cooperative learning method that bring about both individual accountability and achievement of the team goals. In this activity students depend on each other to succeed. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Group members must work together as a team to accomplish a common goal;
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**

The topic is divided into subcategories. Students are divided into groups of five or six students. Then each group would be formed with one student receiving one subcategory of the topic. For this method, each group gets the same set of subcategories. Once individuals have researched and referred their subcategory, they will get together with individuals from the other groups with the same topic to better develop their understanding and become experts of the subcategory. Each student would then return to their original group and teach their subcategory to the rest of their small group.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO8	PO9	PO10	PO12	PSO1
CO5	3	3	2	2	2	2	1

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO8
	3	3	2
Justification for correlation	Basic engineering knowledge is necessary to analyze and characterize the hierarchy of memory. Hence, it is mapped to level 3.	The outcome is mapped to level 3 by identifying the mathematical and engineering knowledge needed	The outcome is mapped to level 2 by demonstrating the types of bus architectures that can be identified by following the



		to solve problems in the memory chip organization.	professional code of ethics while doing activity-based learning.
Innovative practice	PO9	PO10	PSO1
	2	2	1
Justification for correlation	By planning the types of internal communication technologies based on a collaborative activity, Jigsaw helps the students present the results as a team, with smooth integration of contributions from all individual efforts. Hence, the outcome is mapped to level 2.	The topic deals with the design of serial and parallel bus architectures to complement writing and presentations. This task is accomplished through assignments. Hence, it is mapped to level 2.	The students will be able to identify the suitable memory chips for memory allocation. Thus, it is mapped at level 1.

• Images / Screenshot of the practice:



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students gave positive feedback about the activity. They said it was interesting and knowledge improved on the topic. They learn how to communicate with team members and work together.

- ❖ ***Benefit of the practice:***

Every student got equal opportunity to come forward to take part in this activity. The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 85% of students answered correct

- ❖ ***Challenges faced in implementation:***

The main challenge faced is group conflict. Few Students are not ready to work together. To avoid that conflict I started to praise others, taking turns for equal participation, and shared decision making.

References:

- ❖ <https://www.readingrockets.org/strategies/jigsaw>

- ❖ <https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/>


16/12/23

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UNIT IV

1. 08.12.2023 Data Hazards

Exit Slips



Data Hazard:
Data Hazard in pipeline arise when one instruction depends on another instruction (previous ins) result.
eg: add \$s0, \$t0, \$t1
sub \$t2, \$s0, \$t3.
Solution: Forwarding or bypassing.


8/12/23

Signature of Faculty Member



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